

ATROPOS OBLIGATION DAG — FULL MULTI-TIER

Continuation & Completion of Source Document 7 + SD6 Canonical Addenda

Tiers 2 · 3 · 4 + All Deferred Subsystems + Advanced Research Expansions

Maximum atomization · Zero boilerplate · Hybrid Residual RESEARCH+IMPL + HOE Track/WIRE · 2026-10-06

This document is the canonical continuation of the Tier-1 Obligation DAG. It atomizes every deferred domain from Source Document 7 and the SD6 Superiority Addenda (Parts I & II). It is designed to be appended directly beneath Source Document 7.

Extraction rules remain locked: maximum atomization, hybrid format, research prompts force external discovery, impl notes name foreseeable failures and cheapest correct paths that preserve the existing ATROPOS substrate.

TIER-2 — Long-Horizon Governance, Swarm, Storage & Resource Control

Prerequisite: Tier-1 Caller-Completeness Gate (SD7-T1-099) green and a real application has been synthesized end-to-end.

2.1 Strategic Overrides & Architectural Foundations

SD7-T2-001+open-core-airgap-invariant [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research sovereign local-first architectures (Termux, PureOS, GrapheneOS, NixOS offline modes) that keep planning, DAG synthesis and context indexes fully air-gapped while permitting only stateless zero-retention remote inference or CAS storage.

I: Enforce Open-Core Split: core planner, DAG synthesizer, context index and generator remain 100 % open-source local. Hosted plane restricted to optional stateless API or replicated CAS. Air-Gap Invariant: full synthesis and verification operate without network. Foreseeable error: accidental network call inside Wave-0/2; BoundedAgencyGate must refuse any network capability during pure synthesis.

WIRE: BoundedAgencyGate, SourceAuthorityLaw, LocalMemoryStore, CAS

SD7-T2-002+vscode-fork-prohibition [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-001]

R: Research headless daemon + thin surface patterns (LSP servers, Language Server Protocol, headless Claude Code / Aider / Codex) that avoid unmaintainable editor forks.

I: Formal rule: do not fork VS Code. Architecture = local headless daemon (atropos.jar) + thin command surfaces (CLI/TUI + local Bridge HTTP/SSE + Compose Desktop + Android APK + optional thin editor window). Agent process lives outside editor process so it survives crashes, runs in CI and executes headless. Foreseeable error: embedding heavy editor process; keep daemon independent.

WIRE: MainActivity (Android), BridgeHttp, CLI entrypoints, atropos.jar packaging

SD7-T2-003+integration-resequencing [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-001]

R: Research phased integration ladders used by production agent systems that keep local toolchains as Tier-0 and treat chat/issue trackers as distribution, not core.

I: Resequenced tiers: Tier-0 = Filesystem sandbox + local Git + local toolchains (kotlinc, cargo, tsc, go, gradlew). Tier-1 = Sentry (stack→file:line→territory patch→evidence) + GitHub Actions CI (headless clean-machine reproducibility). Tier-2 = Linear/Jira/Confluence/Datadog/Puppeteer. Tier-3 = Slack/Discord/Community MCP. All integrations under BoundedAgencyGate + SecretSinkMatrix + Territory. Raw tokens never enter prompts. Foreseeable error: elevating chat into production core; refuse.

WIRE: BoundedAgencyGate, SecretSinkMatrix, TerritoryGrant, CI configuration

2.2 Swarm Hierarchy, Coordination & Model Routing

SD7-T2-010+six-level-swarm-topology [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-060]

R: Research hierarchical multi-agent topologies (AutoGen, CrewAI, MetaGPT, military command nets) that keep communication $O(N)$ rather than $O(N^2)$ and enforce territory disjointness.

I: Implement 6-level hierarchy: L1 Human Owner/CEO (goal, irreversible approvals, governance mutation); L2 Director Agent (topological decomposition, territory assignment, worktree monitoring, drift detection); L3 Division VP (domain coordination synthesis/verification/storage/delivery); L4 Supervisor Manager (team territory tracking, conflict resolution); L5 Specialist Agent (narrow domain); L6 Worker Agent (isolated atom, zero sibling visibility). Cross-cutting: HR Information Router (audited filtered chokepoint), Auditor (read-only, no execution capture), Custodian (state hygiene, GC, zero LLM overhead). Foreseeable error: flat peer communication; ban and force hierarchical routing.

WIRE: AutonomousWaveOrchestrator, DirectorAgent (new), TerritoryGrant, IsolatedWorktreeService

SD7-T2-011+heterogeneous-provider-matrix [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-010]

R: Research specialized model routing matrices that assign high-velocity drafting, large-context planning, architectural audit, algorithmic cores and zero-cost local validation to distinct providers while surviving 429/offline.

I: Dispatch matrix: Groq Llama-3.3-70B = high-velocity code workers & AST chunk drafting; Gemini 2.5 Pro = large-context spec & DLOI topological planning; Anthropic Claude 3.7 Sonnet = architectural audit, Wave-0 contracts, complex reflexive repair; DeepSeek-V3 = algorithmic routines, heuristic optimization, data structures; Ollama local ARM64 = zero-cost on-device AST validator & security gatekeeper. Fallback: 429 or offline → next provider or local Ollama without pipeline interrupt. Foreseeable error: hard-coding a single provider; RoutePolicy must remain pure data.

WIRE: ProviderCascadeRouter, RoutePolicy, FreeModeGuard

SD7-T2-012+swarm-parallelism-math [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-010]

R: Research Amdahl-derived effective parallelism formulas that subtract coordination overhead, rework conflict and verification debt. Examine real measurements from large agent swarms.

I: $\text{Speedup}(N) = 1 / ((1-p) + p/N)$. $N_{\text{effective}} = N \times (1 - \text{coordinationOverhead}) \times (1 - \text{reworkConflict}) \times (1 - \text{verificationDebt})$. Max parallel lanes = 64 concurrent sub-agents across disjoint worktrees. Critical-path scheduling prioritizes topological bottleneck atoms. $\text{ConflictRisk} \propto \Sigma(|T_i \cap T_j| / |T_i \cup T_j|)$ must remain 0 by territory disjointness. Foreseeable error: optimistic N without measuring overhead; instrument and feed back into $N_{\text{effective}}$.

WIRE: DirectorAgent, PartitionClusterManager, TerritoryGrant

SD7-T2-013+bounded-agency-proposal-gate [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-001, SD7-T2-010]

R: Research strongly-typed action proposal pipelines that never execute raw LLM prose or shell commands ambiently (capability systems, object-capability security, sel4-style enforcement).

I: Flow: Model output → untrusted text → deterministic parser → ActionProposal → BoundedAgencyGate → CapabilityEnforcer → TerritoryEnforcer → sandbox execution. No raw LLM prose or shell commands execute ambiently. All actions strongly typed. Foreseeable error: stringly-typed commands; reject any proposal that is not a sealed data class.

WIRE: BoundedAgencyGate, CapabilityEnforcer, TerritoryEnforcer, ActionProposal

2.3 Storage Constitution, Resource Governance & Hardware Health

SD7-T2-020+storage-constitution-laws [NEW-SUBSTRATE]

Track: S

dependsOn: []

R: Research local-first retention hierarchies and mark-and-sweep GC used by CAS systems (IPFS, Perkeep, Content Addressable Storage in Bazel) that keep phone flash from filling during multi-week autonomous runs.

I: Laws: (01) Global device ceiling $\text{ATROPOS_TOTAL_LOCAL_BYTES} \leq N$ (per-device profile). (02) Retention tiers $\text{HOT} = \text{current checkpoint} + \text{active proposal}$, $\text{WARM} = \text{last } K \text{ successful promotions}$, $\text{COLD} = \text{remote R2 CAS only}$, $\text{EXPIRE} = \text{failed candidates} + \text{rejected proposals} + \text{old worktrees}$. (03) Artifact GC atoms: GC worktrees after promote/reject, GC candidate JARs not in active promotion, cap evidence bundle size (hashes local, bulk on R2). (04) App-factory offload: source+tiny evidence local, full git history and build products to GitHub/R2, local build dirs ephemeral. (05) Pre-flight free-space gate = blocking gate equal to VerifiedCompletionGate. (06) Memory compression: observations become structured bounded records, raw transcripts ephemeral. Foreseeable error: unbounded evidence growth; Custodian must enforce after every wave.

WIRE: Custodian, LocalMemoryStore, CAS, IsolatedWorktreeService, EvidenceCollector

SD7-T2-021+merkle-incremental-build-cache [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T2-020]

R: Research Merkle digest trees for incremental builds (Bazel remote cache, Nix store, content-addressed compilation) that keep rebuild cost $O(\log N)$ at 1 M LOC.

I: Every file has SHA-256. $\text{ModuleHash} = H(\text{Dependencies}) \cup H(\text{Contracts}) \cup H(\text{SourceFiles})$. If ModuleHash unchanged, reuse existing verified binary and skip recompilation. Foreseeable error: hash collisions or partial writes; always verify full content after cache hit.

WIRE: NativeToolchainRegistry, GovernedCompileGate, CAS

SD7-T2-022+thermal-hardware-pacing [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T2-020]

R: Research host-aware throttling for battery, temperature and free RAM used by mobile build systems and long-running Termux agents.

I: Monitor battery level, battery temperature, CPU throttling, free RAM. Throttle law: if $\text{temp} > 42^\circ\text{C}$ or $\text{battery} < 20\%$ reduce parallel workers and pause heavy builds. Ensures safe multi-week execution on pocket hardware. Foreseeable error: ignoring thermal signals; treat as hard gate equal to free-space gate.

WIRE: DirectorAgent, PartitionClusterManager, HostHealthMonitor (new)

SD7-T2-023+mathematical-termination-potential [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-060]

R: Research potential functions and oscillation pruning used in formal verification and search algorithms that guarantee termination of autonomous loops.

I: $\Phi(\text{Goal}) = \text{RemainingUnimplementedAtoms} + \text{OpenVerificationGaps} + \text{UnresolvedDiagnostics}$. Transition accepted iff $\Phi(t+1) < \Phi(t)$. If same error seen twice, trigger hypothesis disputation and prune search branch. Prevents infinite autonomous loops by construction. Foreseeable error: non-monotonic progress; log Φ and halt on non-descent.

WIRE: AutonomousWaveOrchestrator, DirectorAgent, VerifiedCompletionGate

SD7-T2-024+quota-economic-governance [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-011]

R: Research inference cost ledgers, free-mode guards and emergency paid gates that prevent quota collapse while preserving local-first preference.

I: Integrate QuotaLedger + FreeModeGuard + EmergencyPaidGate + RoutePolicy. Cascade: local-first → free-ready provider → free-fallback provider → cooldown queue → offline degraded mode → paid emergency only by explicit unlock. 429 causes cooldown not abort. Continuous quality-per-dollar arbitrage. Foreseeable error: silent paid spend; EmergencyPaidGate requires explicit human unlock.

WIRE: ProviderCascadeRouter, RoutePolicy, QuotaLedger (new), FreeModeGuard

SD7-T2-025+exact-hash-restart-continuity [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T2-020]

R: Research exact-hash restart vectors used by reproducible builds and long-running agent systems that never rebuild goals from model intuition.

I: Restore vector = [RootHumanAuthorityHash, ActiveSourceSpecHash, GoalInvariantFingerprint, DerivationGraphVersion, CurrentAtomId, ActiveTerritory, ActiveContextManifestHash, RemainingBudget, AcceptedFindings, LastVerifiedGate]. Prohibition: never summarize old goal from memory or rebuild from current model intuition. Divergence = blocking halt. Foreseeable error: partial restore; require full vector or refuse.

WIRE: RestartCoordinator, LocalMemoryStore, CAS, SourceAuthorityLaw

TIER-3 — CLI / UNIX Sovereignty + Advanced Anti-Fake-Green Oracles

Prerequisite: Tier-2 storage constitution and swarm topology green.

3.1 Full CLI / UNIX Sovereignty Substrate

SD7-T3-001+pty-process-hosting [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-010]

R: Research isolated POSIX pseudo-terminal (openpty/forkpty) patterns used by tmux, screen, and modern agent CLIs that prevent interactive hangs on npm/cargo/gradle prompts.

I: All external toolchains spawned via isolated PTY. Features: raw termios control, non-blocking I/O poll, max drain timeout 600 000 ms, isolated process group detachment. Foreseeable error: interactive hang; always supply non-interactive flags and kill process group on timeout.

WIRE: BoundedProcessRunner, ToolchainExecutionDriver, all LanguageToolchainDrivers

SD7-T3-002+posix-signal-lifecycle [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-001]

R: Research typed signal handlers that preserve CAS index and Git HEAD integrity under SIGINT/SIGTERM/SIGWINCH/SIGHUP.

I: Handlers: SIGINT → clean worktree restore + record abort evidence; SIGTERM → emit checkpoint CAS snapshot + halt workers; SIGWINCH → dynamic viewport recalculation, no state loss; SIGHUP → background daemon detach. Invariant: signals never corrupt CAS index or Git HEAD. Foreseeable error: unclean shutdown; always write evidence before exit.

WIRE: IsolatedWorktreeService, CAS, RestartCoordinator, Custodian

SD7-T3-003+universal-shell-completion [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research static generation of shell completion from canonical command registries and AST symbol graphs (Click, Cobra, Clap, complete-alias patterns).

I: Generate dynamic flag/argument/file-path/atom autocomplete for Bash, Zsh, Fish, PowerShell directly from CommandRegistry + AstSymbolGraph. No third-party tools. Foreseeable error: stale completions; regenerate on every CommandRegistry change.

WIRE: CommandRegistry, AstSymbolGraph, CLI packaging

SD7-T3-004+cli-tty-discipline [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research strict TTY discipline standards (NO_COLOR, CLICOLOR_FORCE, TERM=dumb, isatty detection) and POSIX exit-code conventions used by professional CLIs.

I: Conformance: NO_COLOR, CLICOLOR_FORCE, TERM=dumb fallback, isatty detection. Output separation: stdout = pure machine JSON/NDJSON or diffs; stderr = telemetry/logs/spinners/diagnostics. POSIX exit codes: 0=verified success, 64=usage/syntax, 65=contract verification failure, 69=unavailable service, 70=internal core fault, 74=IO/storage failure, 77=permission/territory refusal, 78=configuration/policy violation. Foreseeable error: color or progress on stdout; never.

WIRE: CLI entrypoints, all diagnostic emitters

SD7-T3-005+terminal-input-defense [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research bracketed-paste mode and paste-size collapsing used by modern terminals to prevent carriage-return injection and terminal hijacking.

I: Enable ESC[?2004h. Collapse pastes >20 lines into bounded CAS handles. Prevents CR injection and terminal hijacking. Cursor/keyboard: VT100 escapes + UTF-8 raw key decoder + searchable ring buffer. Foreseeable error: accepting multi-line paste as commands; always collapse.

WIRE: CLI input loop, CAS

SD7-T3-006+osc52-clipboard-bridge [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research OSC-52 ANSI clipboard streaming and native fallbacks (wl-copy, xclip, pbcopy, Termux) with secret scrubbing.

I: Protocol: OSC-52 emission. Fallback cascade: Termux clipboard → wl-copy → xclip → pbcopy. Invariant: raw secrets and auth tokens filtered before clipboard write. Foreseeable error: secret leakage; always run SecretSinkMatrix before write.

WIRE: SecretSinkMatrix, CLI surfaces

SD7-T3-007+standalone-virtual-pager [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T3-004]

R: Research zero-dependency terminal pagers with half-page scroll, regex search and unified-diff syntax highlighting.

I: Internal zero-dependency pager: forward/backward half-page scroll, regex search, unified-diff syntax highlighting, non-blocking escape. Foreseeable error: external less/more dependency; keep pure internal.

WIRE: CLI surfaces, diagnostic viewers

3.2 Advanced Anti-Fake-Green & Anti-Boilerplate Oracles

SD7-T3-010+property-based-fuzzing-oracle [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-030, SD7-T1-072]

R: Research property-based fuzzing across extreme integers, Unicode stress strings, recursive JSON structures, null-pointer injections and race-condition concurrency (Hypothesis, QuickCheck, cargo-fuzz patterns).

I: Automated fuzz test cases across type domains. Predicate: module must satisfy formal invariants across 1 000 deterministic PRNG iterations. Foreseeable error: non-deterministic seeds; record seed in evidence bundle.

WIRE: Wave-1 test generator, VerifiedCompletionGate, EvidenceCollector

SD7-T3-011+callgraph-reachability-proof [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-071]

R: Research directed AST call-graph traversal that proves every synthesized symbol has an unbroken path from application entry point (main()) or public RPC/HTTP router).

I: Every synthesized class/function/port must have unbroken path from entry point. Isolated orphan symbols without invocation edges trigger immediate build rejection. Zero dead-code tolerance. Foreseeable error: reflection/dynamic dispatch hiding edges; require explicit registration or fail.

WIRE: ArchitectureComplianceChecker, VerifiedCompletionGate, AstSymbolGraph

SD7-T3-012+dual-oracle-differential-compilation [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-010, SD7-T1-061]

R: Research dual-toolchain differential compilation (kotlinc JVM vs GraalVM Native, tsc vs OXC/SWC, go vs go vet -race) that eliminates runtime-specific false greens.

I: Compile and execute suite under two distinct toolchains/modes. Divergent behavior or compiler-specific green fails closed. Foreseeable error: accepting single-toolchain success; always require both oracles.

WIRE: NativeToolchainRegistry, VerifiedCompletionGate, GovernedCompileGate

TIER-4 — Cloud Delivery, Schema Compilation & Autonomous Deployment

Prerequisite: Tier-3 CLI sovereignty and anti-fake-green oracles green.

4.1 Storage Schema Compilation

SD7-T4-001+StorageSchemaCompiler [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T1-020]

R: Research generation of relational/NoSQL DDL, indexes and typed ORM/client mappings that stay bit-identical to Wave-0 contracts (SQLAlchemy, Prisma, Room, Exposed, SQLx patterns).

I: Generate relational/NoSQL DDL, database indexes, typed ORM/client query mappings matching Wave-0 contracts. ATOM-L1-003. Foreseeable error: schema drift from contracts; regenerate from contracts only, never hand-edit.

WIRE: ContractLatticeSynthesizer, SupabaseSchemaDriver, VerifiedCompletionGate

4.2 Cloud Deployment Drivers

SD7-T4-010+CloudDeploymentAdapter [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T1-060]

R: Research uniform non-mock delivery seams that abstract Vercel/Netlify/Supabase/local-dist behind a single contract.

I: Uniform contract for non-mock deployment to hosting providers. No shell scripts that echo success. Foreseeable error: mock deploy that returns fake URL; require real HTTP 200 health-check.

WIRE: AutonomousWaveOrchestrator, all concrete drivers

SD7-T4-011+VercelDeploymentDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T4-010]

R: Research direct REST SHA-1 content-addressed upload to Vercel Deployments API with manifest synthesis and production URL extraction.

I: Direct API deployer packaging build outputs to Vercel production endpoints. SHA-1 content-addressed streaming. Extract live production URL. Foreseeable error: accepting non-200; fail closed until health-check passes.

WIRE: CloudDeploymentAdapter, SecretSinkMatrix

SD7-T4-012+NetlifyDeploymentDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T4-010]

R: Research ZIP REST deployment to Netlify API with production URL extraction.

I: Direct ZIP REST deployer. Same health-check invariant. Foreseeable error: missing site ID or token; fail closed with clear diagnostic.

WIRE: CloudDeploymentAdapter, SecretSinkMatrix

SD7-T4-013+SupabaseSchemaDriver [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-T4-001, SD7-T4-010]

R: Research automated PostgreSQL migration runners with Row-Level Security policy synthesis and connection-pool bounds.

I: Applies SQL migrations, generates RLS policies, registers DB configurations. Enforce connection-pool bounds from Wave-0 contracts. Foreseeable error: open RLS; default deny and require explicit policies.

WIRE: StorageSchemaCompiler, CloudDeploymentAdapter, SecretSinkMatrix

SD7-T4-014+AutonomousDeliveryEngine [NEW-SUBSTRATE]

Track: S

dependsOn: [SD7-T4-010, SD7-T4-011, SD7-T4-012, SD7-T4-013]

R: Research autonomous deployment pipelines that assert HTTP 200 health checks at production domains or bit-verified binary on disk before declaring success.

I: ATOM-L4-003. Autonomous deployment pipeline. Acceptance = HTTP 200 live health-check at target production URL or bit-verified ELF/APK on disk.

Foreseeable error: declaring success on upload alone; always require health-check or binary proof.

WIRE: AutonomousWaveOrchestrator, CloudDeploymentAdapter, VerifiedCompletionGate, EvidenceCollector

SPECGRAPH FOUNDRY — Expanded Ingestion Atoms

These expand the over-condensed SD7-T1-065 into full maximum atomization.

SD7-SG-001+unbounded-ingestion-pipeline [NEW-SUBSTRATE]

Track: B

dependsOn: []

R: Research multi-gigabyte streaming parsers that avoid stack recursion crashes (iterative tokenization, chunked SHACL validation, memory-mapped reading).

I: Concrete mutations to statement_segmentation.py and atomic_decomposition.py for multi-gigabyte spec parsing without stack recursion. Stream rather than materialize. Foreseeable error: recursion depth; convert all recursive descent to iterative.

WIRE: SpecGraphCompiler, atomic_decomposition.py, statement_segmentation.py

SD7-SG-002+SHACL-validation-proof-bundles [NEW-SUBSTRATE]

Track: B

dependsOn: [SD7-SG-001]

R: Research RDF/SHACL shape conformance and feedback-arc-set reduction for formal dependency graphs.

I: Mutations to dependency_compiler.py and shacl_validation.py for RDF/SHACL shape conformance and feedback arc set reduction. Emit formal proof bundles. Foreseeable error: cycles surviving; fail closed on any remaining arc.

WIRE: SpecGraphCompiler, dependency_compiler.py, shacl_validation.py, InternalExecutionDagSynthesizer

ADVANCED RESEARCH EXPANSIONS (Beyond Source Doc 7)

These atoms advance the architecture with 2026-era techniques discovered by external research pressure.

SD7-ADV-001+capability-based-security-model [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-013]

R: Research object-capability (ocap) and seL4-style capability systems that can be applied to LLM action proposals so that possession of a capability is the only authority.

I: Elevate BoundedAgencyGate to a pure capability system. Every ActionProposal must carry the exact set of capabilities required; missing capability = hard reject.

Foreseeable error: ambient authority; purge any ambient capability.

WIRE: BoundedAgencyGate, CapabilityEnforcer, ActionProposal

SD7-ADV-002+content-addressed-evidence-merkle [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-020, SD7-T2-021]

R: Research pure content-addressed evidence ledgers that make every verification receipt independently verifiable by third parties without trusting ATROPOS.

I: Every evidence bundle is a Merkle tree of hashes. Third-party verifier can validate the Zero-Entropy Proof Certificate in seconds. Foreseeable error: mutable evidence; all evidence must be append-only CAS.

WIRE: EvidenceCollector, CAS, IntegrityProofs, VerifiedCompletionGate

SD7-ADV-003+formal-wave-invariant-checker [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T1-060, SD7-2050-FMC-01]

R: Research static checkers that prove each wave respects the clean-room mutual-information invariant and territory disjointness before any LLM call is issued.

I: Before Wave-2 dispatch, a formal checker proves $I(f_j; f_k | I) = 0$ and Territory disjointness. Failure aborts the wave. Foreseeable error: runtime-only checks; move to pre-dispatch static proof.

WIRE: AutonomousWaveOrchestrator, PartitionClusterManager, ArchitectureComplianceChecker

SD7-ADV-004+provider-quality-arbitrage-ledger [AMENDMENT-CANDIDATE]

Track: S

dependsOn: [SD7-T2-024]

R: Research continuous quality-per-dollar measurement and Pareto routing across heterogeneous providers that updates after every verified atom.

I: After every successful VerifiedCompletionGate, record (provider, token cost, mutation-kill score, SMT discharge time). RoutePolicy continuously re-ranks. Foreseeable error: stale rankings; decay half-life of 48 h.

WIRE: RoutePolicy, QuotaLedger, ProviderCascadeRouter, EvidenceCollector

SD7-ADV-005+deterministic-replay-harness [AMENDMENT-CANDIDATE]

Track: B

dependsOn: [SD7-T2-025, SD7-T3-010]

R: Research deterministic replay of entire multi-week autonomous runs from exact-hash vectors and recorded PRNG seeds so that any failure can be reproduced bit-identically.

I: Every run records the full restore vector + all PRNG seeds + provider response hashes. Replay harness can re-execute any partition offline. Foreseeable error: non-deterministic external services; pin all responses via CAS.

WIRE: RestartCoordinator, EvidenceCollector, CAS, IsolatedWorktreeService

CROSS-TIER ACCEPTANCE GATES

SD7-GATE-T2+tier2-completion [NEW-SUBSTRATE]

| Maximum atomization | Hybrid Residual+HOE | All tiers complete

Track: S

dependsOn: [ALL Tier-2 atoms]

R: Research completeness criteria for long-horizon governance layers that must be live before CLI sovereignty is activated.

I: Tier-2 complete when: (1) Open-Core + Air-Gap invariants enforced, (2) 6-level swarm topology live with territory disjointness, (3) Storage Constitution Laws 01–06 enforced by Custodian, (4) Thermal pacing and Φ potential active, (5) Exact-hash restart continuity proven by forced restart test, (6) QuotaLedger prevents silent paid spend. Foreseeable error: partial governance; all six must be green.

WIRE: CI job, atropos doctor --tier2-completeness

SD7-GATE-T3+tier3-completion [NEW-SUBSTRATE]

Track: S

dependsOn: [ALL Tier-3 atoms, SD7-GATE-T2]

R: Research completeness criteria for sovereign terminal + anti-fake-green layers.

I: Tier-3 complete when: (1) All external toolchains run under PTY with signal lifecycle, (2) Shell completion generated for four shells, (3) TTY discipline and exit codes enforced, (4) Property-based fuzzing, call-graph reachability and dual-oracle differential all blocking, (5) Zero orphan symbols in a synthesized application. Foreseeable error: accepting single-toolchain success; dual oracle mandatory.

WIRE: CI job, atropos doctor --tier3-completeness

SD7-GATE-T4+tier4-completion [NEW-SUBSTRATE]

Track: S

dependsOn: [ALL Tier-4 atoms, SD7-GATE-T3]

R: Research end-to-end delivery acceptance that requires live production health-check or bit-verified binary.

I: Tier-4 complete when: (1) StorageSchemaCompiler emits DDL matching Wave-0, (2) At least one cloud driver (Vercel or Netlify) deploys a real application and receives HTTP 200, (3) SupabaseSchemaDriver applies RLS policies, (4) AutonomousDeliveryEngine emits Zero-Entropy Proof Certificate containing deployment URL or binary hash. Foreseeable error: upload-only success; health-check mandatory.

WIRE: CI job, atropos doctor --tier4-completeness, AutonomousDeliveryEngine

CRITICAL PATH ACROSS TIERS

Tier-1 (already delivered) → SD7-GATE-T1 (Caller-Completeness) →

Tier-2 governance + swarm + storage → SD7-GATE-T2 →

Tier-3 CLI sovereignty + advanced oracles → SD7-GATE-T3 →

Tier-4 cloud delivery + schema → SD7-GATE-T4 →

Full system accepted only when all four gates are green and a multi-week autonomous run has produced a Zero-Entropy Proof Certificate.

NON-NEGOTIABLE INVARIANTS (ALL TIERS)

I(Module_A; Module_B | Interface_Lattice) = 0.000 bits

Territory(P_r) ∩ Territory(P_s) = ∅

Non-zero native exit code = terminal rejection

False-Green Tolerance = 0.0000 %

Air-Gap Invariant for pure synthesis and verification

Exact-hash restart continuity; never rebuild goal from model intuition

Every code byte originates from atomic LLM synthesis against formal interfaces verified by genuine native compilers

C_real = min(C_impl, C_int, C_ver, C_ev)

$\Phi(t+1) < \Phi(t)$ monotonic descent or halt

END OF FULL MULTI-TIER OBLIGATION DAG

Designed for direct append beneath Source Document 7 · Maximum atomization of every deferred domain · Advanced research expansions included · 2026-10-06